Prizmatic

System Design Presentation

Architecture | Technology Choices | Data Design | Sequence Diagrams | Project Schedule

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Problem & Solution

THE CHALLENGE

Small software teams often lack a dedicated project manager, forcing developers to manually manage tasks, deadlines, progress tracking, and pull request coordination across multiple tools.

OUR SOLUTION

Prizmatic provides a centralized project management dashboard with an AI project manager that helps assign tasks, recommend priorities, track deadlines, monitor pull requests, and reduce coordination overhead.

Requirement & Design Updates

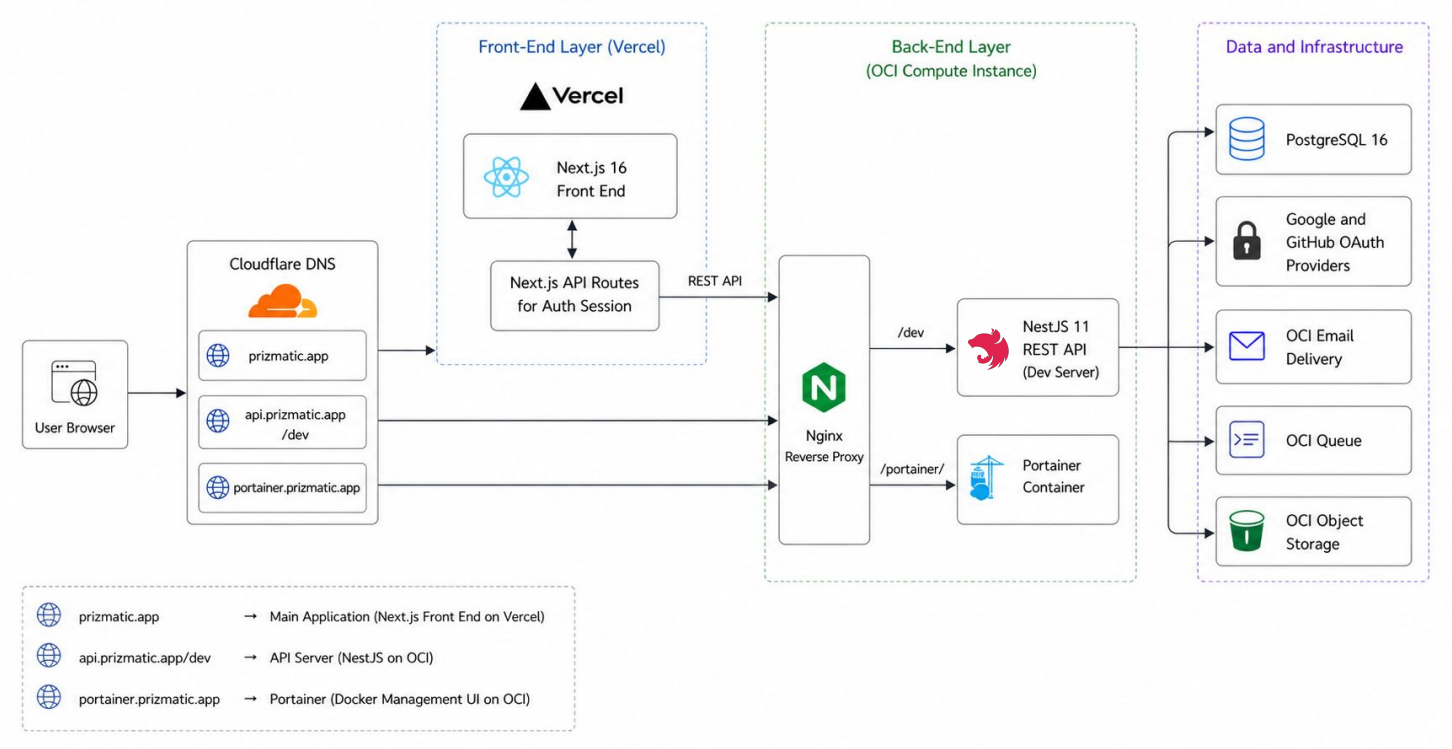
Workspace Management	Invitations, update/delete workspace
Project Design	Belongs to workspace, project-specific members, project member job assignments
Work Items	Hierarchy, type, assignees, labels, priority, workflow status

Technology Choices

Layer	Technology	Reason
Frontend	Next.js 16, React 19, TypeScript	Modern UI, routing, SSR/CSR, type safety
UI & State	Tailwind CSS 4, shadcn UI, React Query 5, Zustand 5	Consistent UI, accessible components, caching, state
Backend	NestJS 11, Node.js 22, TypeScript	Modular REST API, DI, guards, interceptors
Database	PostgreSQL 16, TypeORM 0.3	Relational data, joins, transactions, migrations
Auth	JWT, Argon2, Helmet, rate limiting	Secure authentication, hashing, API protection
Infra	Docker, OCI services, GitHub Actions	Repeatable deploy, email/storage/queue, CI

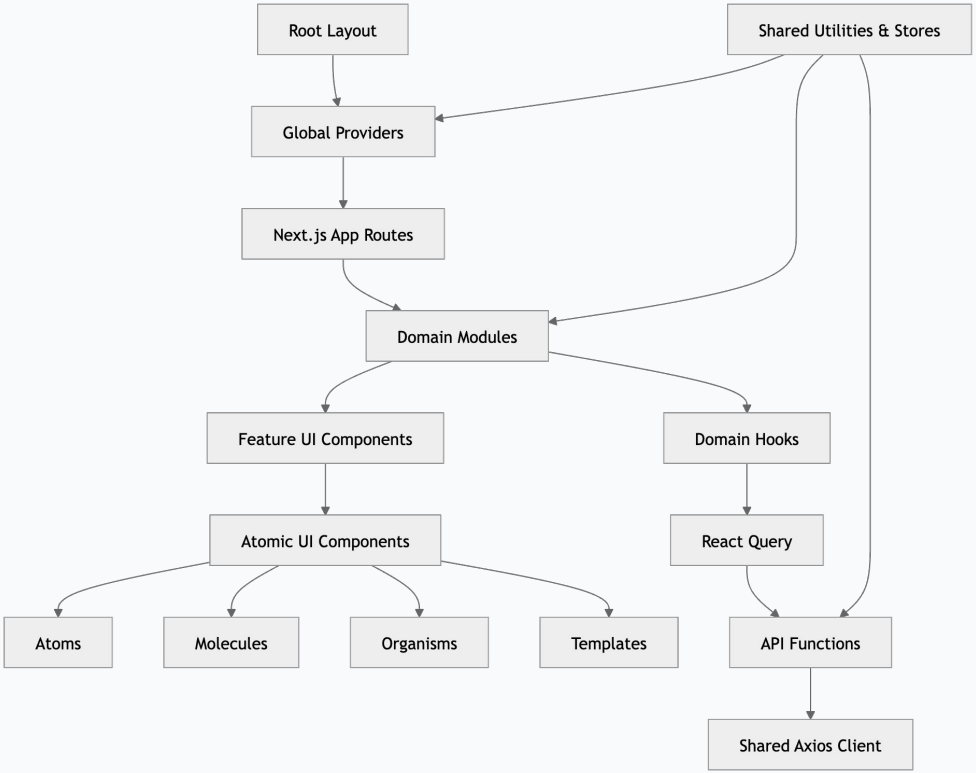
TypeScript across the full stack reduces context switching and ensures consistent API contracts.

High-Level Architecture



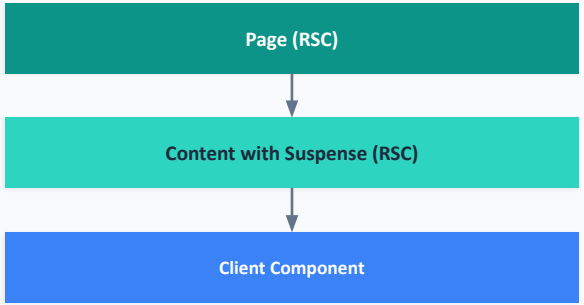
External services are accessed only through the backend layer.

Frontend Architectures

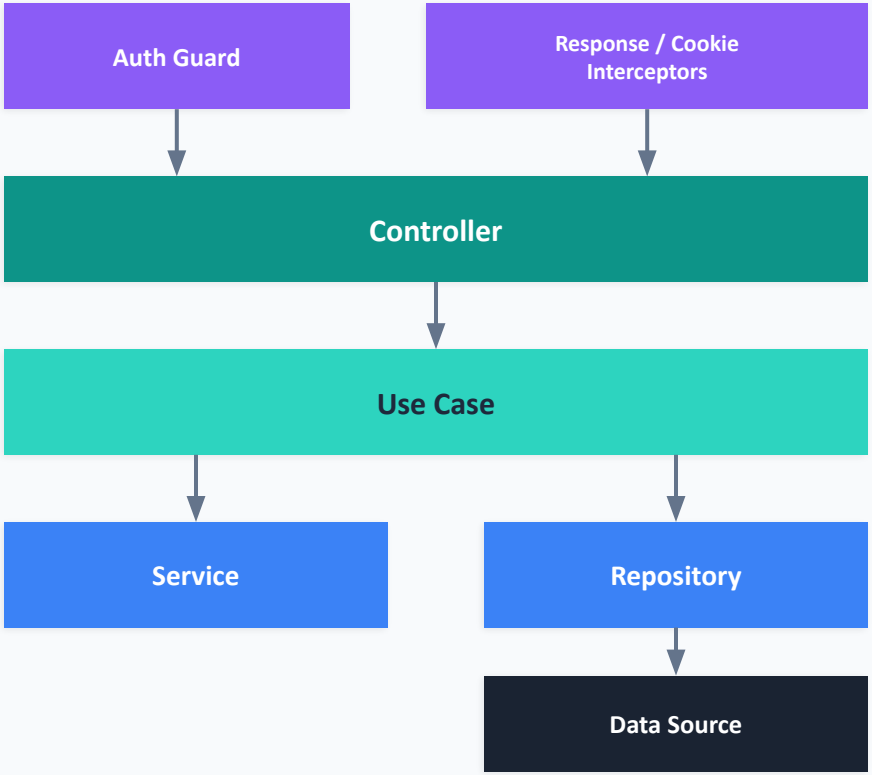


KEY PATTERNS

- Domain-driven Design
- Atomic Design
- Prefetching (SSR)
- Tanstack Query for Caching
- Request Proxy



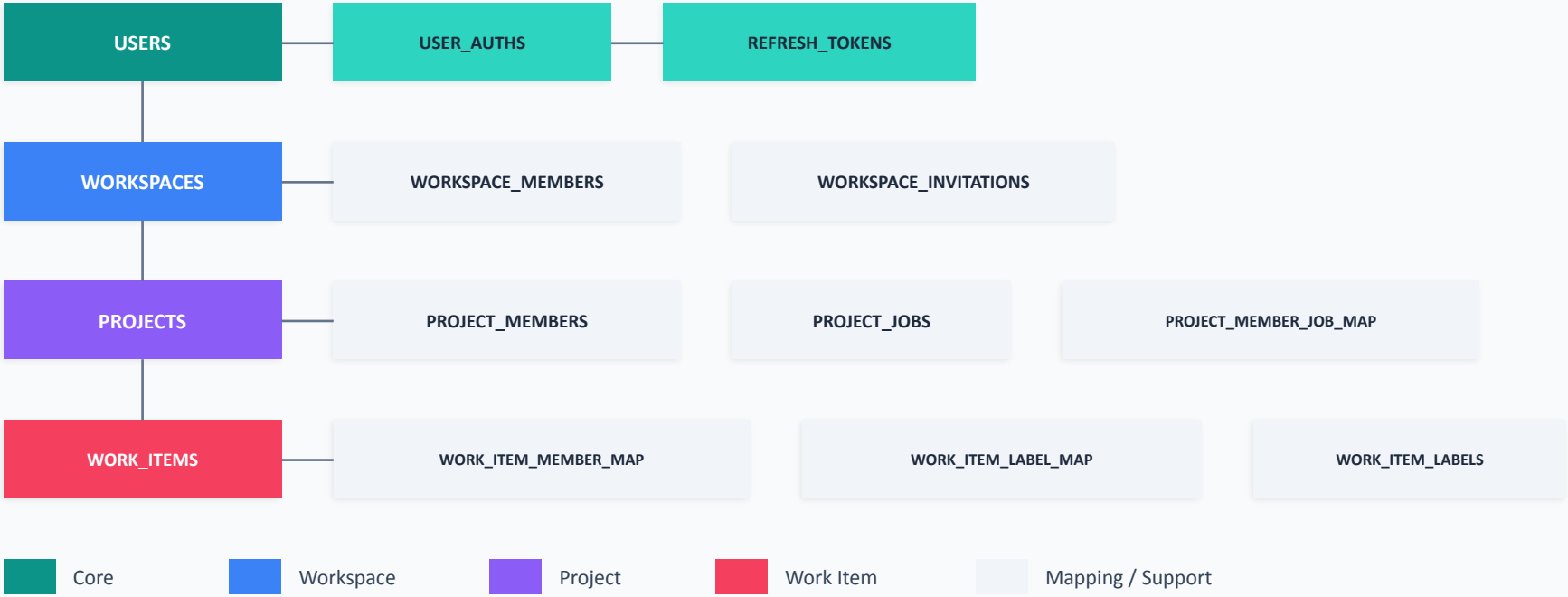
Backend Architectures



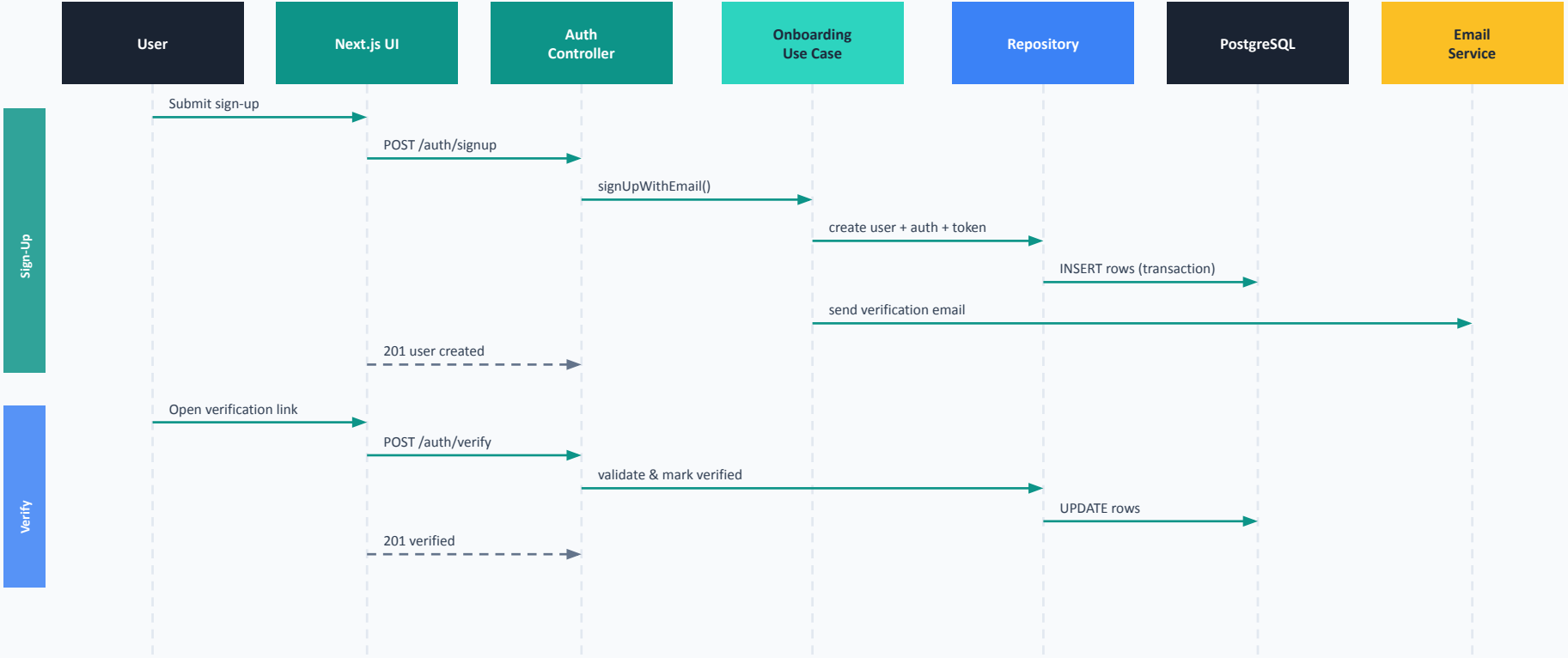
KEY PATTERNS

- Domain-driven Design
- Clean Architecture
- Repository pattern
- Dependency injection
- DTO-based validation
- Unit of Work for transactions

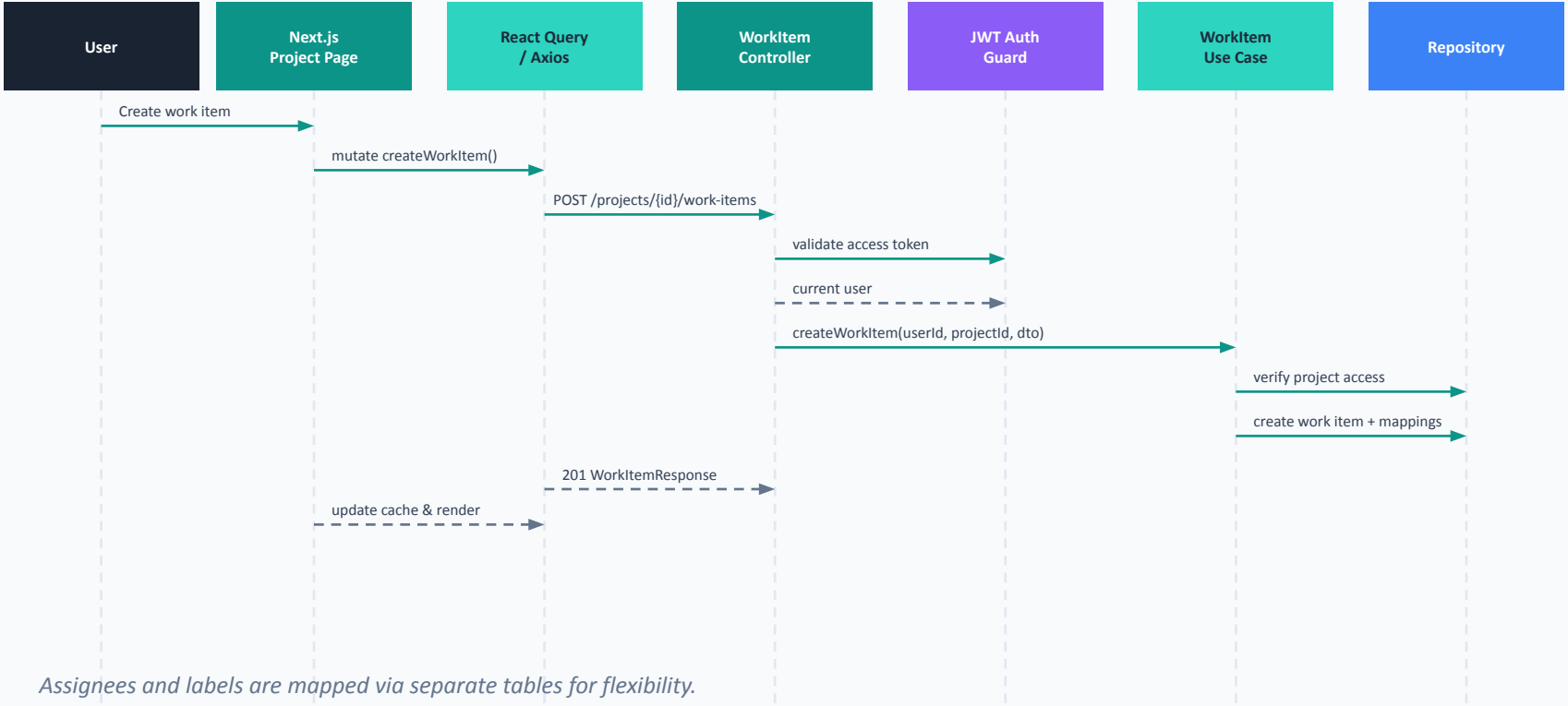
Simplified Data Design



Email Sign-Up & Verification Flow



Create Work Item Flow



Assignees and labels are mapped via separate tables for flexibility.

Project Schedule

Week 9	**Publishing Workspace Features Pages**	Workspace CRUD, members, roles, invitations
Week 10	**Publishing Project/Work Item Features Pages**	Project CRUD, members, job roles, assignments
Week 11	**Publishing Comments and Artifacts Pages**	Comments, artifact links, attachments, work item discussions
Week 12	**AI Agent Integration**	AI integration, priority recommendations, task suggestions, PR feedback
Week 13	**Testing & Polish**	Bug fixes, UI polish, API cleanup
Week 14	**Final Release**	Final deployment, documentation, demo prep

Prizmatic

Modular TypeScript full-stack architecture supporting secure authentication, workspace collaboration, project management, and work item tracking.

DESIGN PRIORITIES

- Maintainability
- Clear API Contracts
- Relational Data Integrity
- Realistic Deployment Path

Thank You — Questions?